Global NCP Analysis: Data Dictionary for Output Tables

This document provides detailed definitions for the key CSV output tables generated by the analysis pipeline. These tables contain the core quantitative results for hotspot intensity, multi-service overlap, socioeconomic profiling, and driver attribution.

`hotspot_area_stats.csv`

- **Source**: `analysis/hotspot_synthesis.qmd`
- **Purpose**: Quantifies hotspot intensity, global share, and relative burden by geographic or socioeconomic groupings.

Column	Meaning
`service`	The ecosystem service being analyzed.
`group`	The specific value of the grouping variable (e.g., 'East Asia & Pacific', 'Low income').
`n_total`	Total number of valid grid cells within that `group`.
`n_hot`	Number of hotspot cells within that `group`.
`pct_area`	**Hotspot Intensity (Coverage)**. The percentage of the group's area that is a hotspot. Formula: `(n_hot / n_total) * 100`.
`global_n_hot`	The total number of hotspots for that service across the entire globe.
`pct_share`	**Global Share**. The percentage of all global hotspots for a service that are located within this `group`. Formula: `(n_hot / global_n_hot) * 100`.
`grouping_var`	The grouping variable used for the aggregation (e.g., 'region_wb', 'income_grp').
`expected_share`	The group's percentage of the total global land area. This is the share of hotspots the group *would* have if hotspots were distributed evenly.

Column	Meaning
`relative_intensity`	**Enrichment Score**. A measure of disproportionate burden. It is the ratio of the group's actual share of hotspots to its expected share. A score > 1 indicates the region is a "hotspot of hotspots." Formula: `pct_share / expected_share`.

`hotspot_multiservice_stats.csv`

- **Source**: `analysis/hotspot_synthesis.qmd`
- **Purpose**: Quantifies the overlap of multiple service hotspots ("Hotness") within geographic or socioeconomic groupings.

Column	Meaning
`group`	The specific value of the grouping variable.
`mean_hotspots`	**"Hotness"**. The average number of overlapping service hotspots per grid cell within that group.
`pct_multi`	The percentage of grid cells within the group that have more than one hotspot.
`n_pixels`	The total number of grid cells in the group.
`grouping`	The grouping variable used for the aggregation.

`hotspot_pop_exposure.csv`

- **Source**: `analysis/hotspot_synthesis.qmd`
- **Purpose**: Quantifies the absolute number of people living within hotspot areas.

Column	Meaning
`service`	The ecosystem service being analyzed.
`income_grp`	The World Bank Income Group of the grid cell.

Column	Meaning
`hdi_bin`	The Human Development Index (HDI) bin of the grid cell.
`exposed_population`	The total population (from GHS-POP) living within the hotspot cells for that specific service and socioeconomic group.
`hotspot_cells`	The number of hotspot cells corresponding to that service and group.

`ks_results_hot_vs_non.csv`

- **Source**: `analysis/KS_tests_hotspots.qmd`
- **Purpose**: Provides the statistical results from the Kolmogorov-Smirnov tests comparing the socioeconomic profiles of hotspot vs. non-hotspot cells.

Column	Meaning
`service`	Ecosystem service being analyzed.
`var`	Socioeconomic variable being tested (e.g., GDP, Population).
`D`	The Kolmogorov–Smirnov D statistic, representing the maximum absolute difference between the empirical cumulative distribution functions (ECDFs) of the hotspot and non-hotspot groups. Larger values indicate a greater difference.
`p_value`	Two-sided KS p-value. Small values (e.g., < 0.05) indicate the two distributions are significantly different.
`cliffs_delta`	A non-parametric effect size (`2*auc - 1`) that quantifies the magnitude and direction of the difference. Ranges from -1 (all hotspot values are smaller) to +1 (all hotspot values are larger). A value of 0 means the distributions are perfectly overlapping.
`mean_hot`, `mean_non`	Mean of the variable in the hotspot and non-hotspot groups.
`median_hot`, `median_non`	Median of the variable in the hotspot and non-hotspot groups.

Column	Meaning
`p_adj`	Benjamini–Hochberg FDR-adjusted version of `p_value` (corrected for multiple comparisons).

(Note: See the full data dictionary in `analysis/KS_tests_hotspots.qmd` for additional statistical columns).

`lcc_es_hotspot_overlap_... .csv`

- **Source**: `analysis/hotspot_extraction.qmd`
- **Purpose**: Quantifies the "Attribution Gap" by measuring the spatial overlap between Ecosystem Service (ES) hotspots and Land Cover Change (LCC) hotspots.

Column	Meaning
`service`	The ecosystem service being analyzed.
`n_es_hotspots`	Total number of hotspots for that ecosystem service.
`n_overlap`	Number of ES hotspots that are also Land Cover Change hotspots.
`pct_overlap`	**The Attribution Metric**. The percentage of ES hotspots that spatially overlap with LCC hotspots. A low percentage indicates a large "Attribution Gap."
`driver`	The specific land cover change driver being tested (e.g., `Forest_Loss`, `Urban_Exp`, `Crop_Exp`).
`metric`	The change metric used for the hotspot definition (`pct_chg` or `abs_chg`).

Granular Land Cover Change Metrics

- **Source**: `analysis/LC_change_granular.qmd`
- **Purpose**: These columns, found in `processed/10k_lcc_granular_metrics.gpkg`, define the specific land cover drivers used in the attribution analysis.

Column Name	Interpretation
`ForestLoss_Loss_Forest_1992_2020`	**Gross Deforestation**. The specific area of Forest that converted to Non-Forest.
`Expansion_Gain_Urban_1992_2020`	**Urbanization**. The area converted to Urban from any other class.
`Expansion_Gain_Cropland_1992_2020`	**Agricultural Expansion**. The area converted to Cropland from any other class.
`GrasslandLoss_Loss_Grassland_1992_2020`	**Grassland Loss**. The specific area of Grassland that converted to another class.
`ForestLoss_dir_ch_Forest_1992_2020`	**Net Forest Change**. (Gain - Loss). Negative values indicate net deforestation.